Business overview

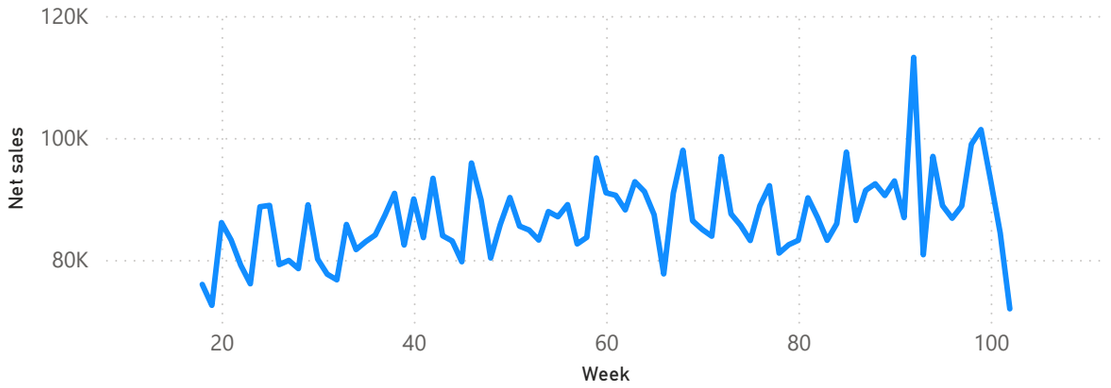
dunnhumby The Complete Journey: 2,500 households, 102 weeks. Weeks 1-17 (households still joining the panel) are filtered out.

Weeks

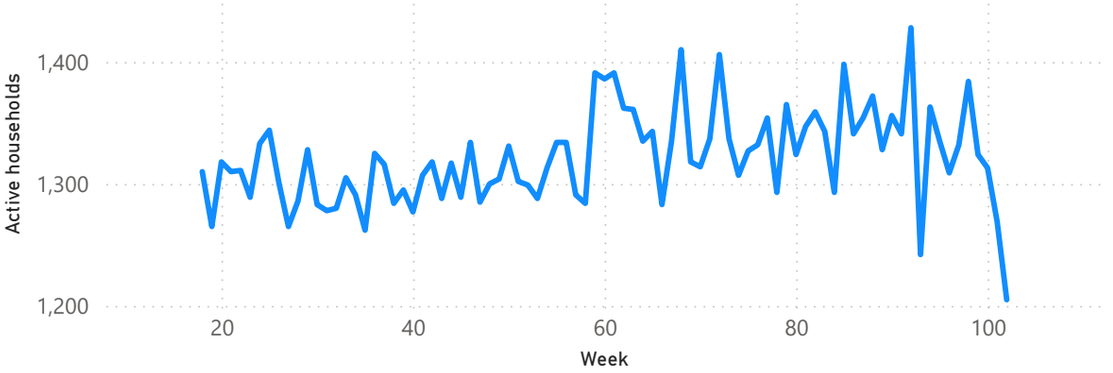
18 102

7.38M	1,318	29.33	89.8%
Net sales	Active households (median per week)	Basket value	4-week retention

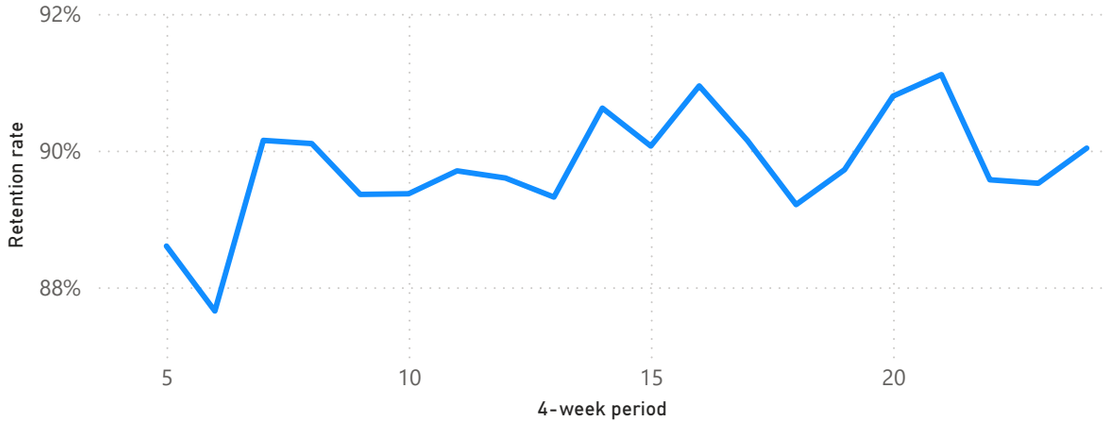
Net sales per week



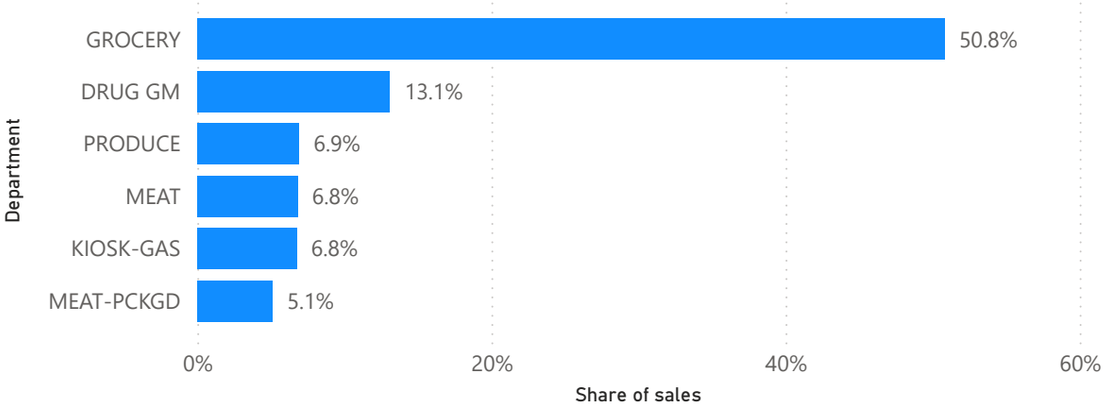
Active households per week



Retention: active again in the next 4-week period



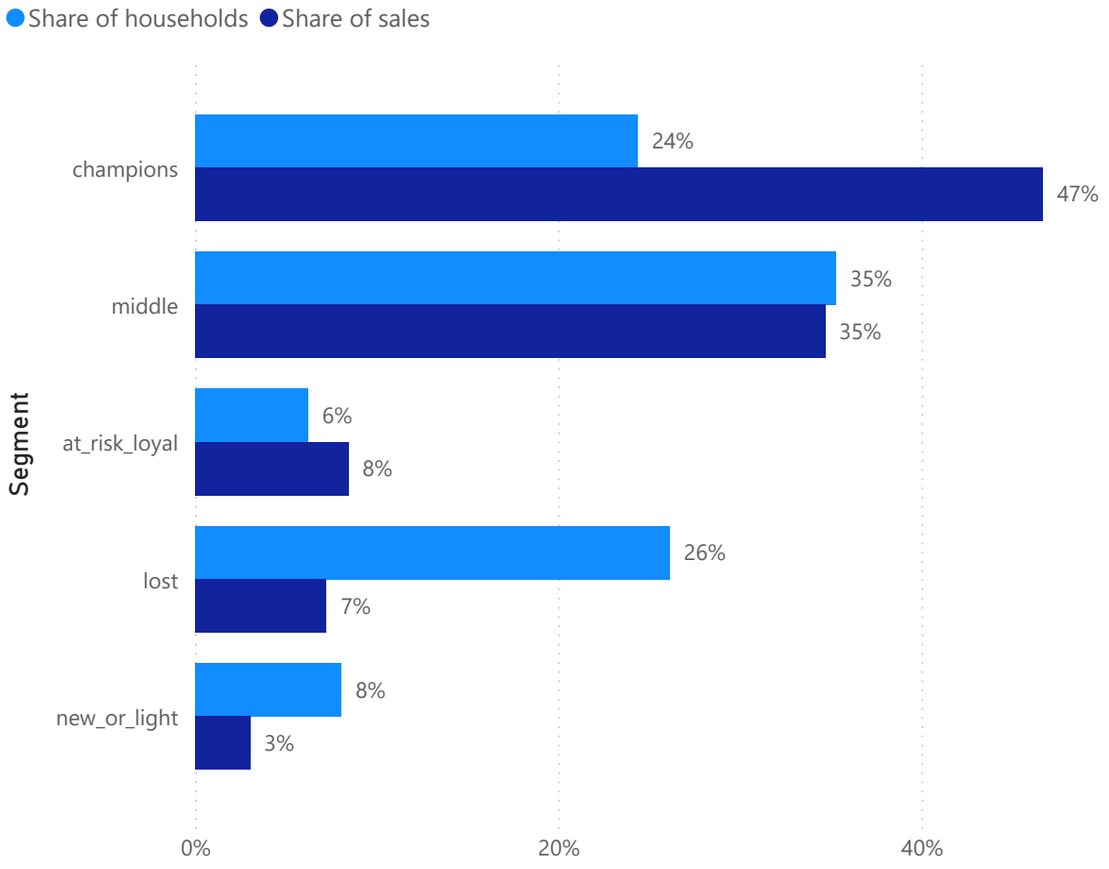
Top 6 departments by share of sales (all weeks)



Customer segments (RFM)

Champions are 24% of households but bring 47% of sales. Lost households are 26% of households and only 7% of sales.

Share of households vs share of sales



Segment table (weeks are medians)

Segment	Households	Sales / household	Active weeks	Weeks since last buy
champions	609	6,173	77	0
at_risk_loyal	155	4,380	67	2
middle	882	3,168	51	1
new_or_light	201	1,209	28	0
lost	653	891	22	5
Total	**2,500**	**3,223**		

How to read it

RFM = recency, frequency and monetary value over the whole 102 weeks (sql/02_rfm.sql). The segments describe customers; the churn model does not use them, because they contain the future.

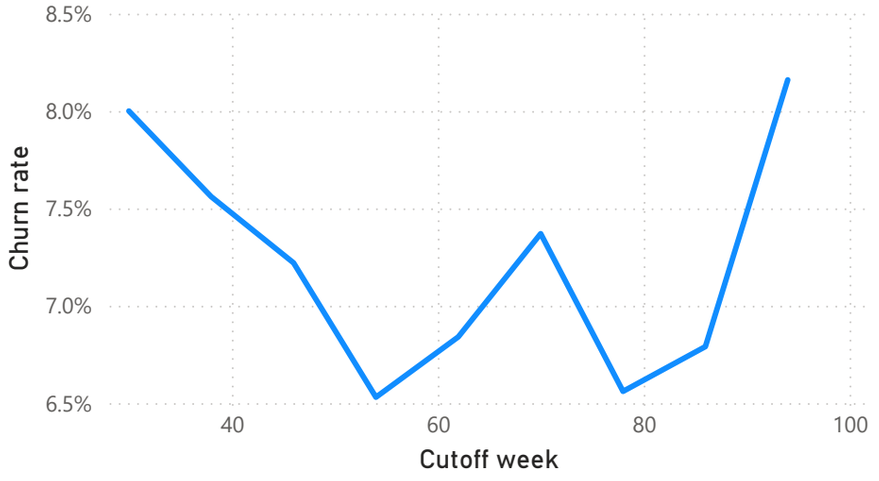
Losing one champion costs far more than losing one light shopper: the coupon list ranks households by expected sales lost, not by churn risk alone.

Churn & coupon list

Churn = no purchase in the next 8 weeks, among households that bought in the last 12 weeks. Coupon list = the 10% of at-risk households with the highest expected sales lost in week 102.

8.2%	+1.4	231	8,194
Churn rate (week 94)	Change vs week 86 (percentage points)	Households on the coupon list	Expected sales lost on the list

8-week churn rate per snapshot



Coupon list (highest expected sales lost first)

Household	Churn risk	Sales (12 wk)	Expected loss	Weeks silent	Active weeks (12 wk)
149	18.3%	753	138.2	8	4
1016	23.0%	508	116.6	8	2
2042	20.0%	467	93.2	4	2
1629	40.3%	228	92.1	7	1
1702	21.3%	402	85.5	8	3
910	29.5%	285	84.1	10	1
2420	15.7%	485	76.0	6	5
2251	22.3%	340	75.9	7	2
2374	13.1%	520	68.1	4	5
1215	22.1%	307	67.8	2	2
1140	30.2%	215	65.0	7	2
700	5.8%	1,092	63.3	1	7
1205	17.7%	349	61.9	4	7
1657	20.3%	297	60.1	4	4
2345	5.4%	1,116	60.0	2	9
1196	13.2%	453	59.8	4	4
2136	7.3%	805	58.6	1	6
931	20.5%	284	58.3	6	2
1496	37.4%	153	57.4	7	1
237	41.2%	136	55.9	8	1
1507	8.1%	690	55.9	1	7

How to use the list

Expected sales lost = churn probability x sales in the last 12 weeks (src/predict.py).
Whether a coupon really keeps a household is unknown: test it first. Give the coupon to a random half of the list and compare churn after 8 weeks (A/B test).